

PAULATUK, NT, Canada

WMO: 719840

Lat: 69.360N

Lon: 124.080W

Elev: 21

StdP: 14.68

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.3	-32.5	-42.8	0.5	-34.0	-39.4	0.6	-32.2	47.8	0.4	42.6	0.3	5.4	240	0.992

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	71.5	58.9	66.7	56.8	62.4	54.9	60.1	69.6	57.7	65.5	55.2	61.8	11.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.6	65.9	63.7	53.4	60.7	61.5	51.2	55.9	58.9	26.6	70.3	24.9	65.4	23.4	61.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.4	29.1	24.8	DB	-38.9	79.4	4.9	4.4	-42.4	82.6	-45.3	85.2	-48.0	87.6	-51.6	90.8	(4)
(5)			WB	-39.0	64.1	5.1	3.3	-42.6	66.5	-45.6	68.4	-48.5	70.2	-52.2	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	-12.0	-12.5	-10.3	6.3	26.7	41.8	50.6	47.8	38.7	22.6	4.9	-7.9	(6)	
	DBStd	25.71	12.04	12.17	13.22	13.08	10.70	8.15	8.11	7.71	7.41	8.31	10.25	11.76	(7)	
	HDD50	12427	1923	1749	1870	1311	723	277	93	137	348	849	1353	1794	(8)	
	HDD65	17691	2388	2169	2335	1761	1188	697	452	535	790	1314	1803	2259	(9)	
	CDD50	220	0	0	0	0	1	31	111	69	8	0	0	0	(10)	
	CDD65	10	0	0	0	0	0	1	6	3	0	0	0	0	(11)	
	CDH74	66	0	0	0	0	0	14	35	17	0	0	0	0	(12)	
	CDH80	7	0	0	0	0	0	2	3	2	0	0	0	0	(13)	
	WSAvg	10.0	11.3	11.0	10.0	9.0	9.0	8.2	8.7	9.5	10.8	11.2	11.5	10.2	(14)	
	PrecAvg	8.6	0.4	0.4	0.6	0.6	0.5	0.7	1.0	1.5	0.9	0.8	0.6	0.5	(15)	
(16) Precipitation	PrecMax	13.7	0.9	0.8	2.5	3.7	1.9	1.9	1.7	3.3	2.6	2.9	2.4	1.6	(16)	
	PrecMin	5.7	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.6	0.0	0.0	0.1	0.0	(17)	
	PrecStd	2.5	0.2	0.3	0.6	0.7	0.4	0.5	0.4	0.7	0.5	0.6	0.5	0.4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	20.6	27.5	42.9	57.0	74.7	78.7	76.3	62.8	46.0	33.6	23.2	(19)	
		MCWB	25.1	19.1	25.3	37.9	47.2	60.1	62.8	61.3	54.0	42.4	31.5	22.0	(20)	
	2%	DB	17.9	15.3	21.2	35.8	49.2	67.2	72.8	68.9	56.8	37.6	27.8	18.0	(21)	
		MCWB	16.6	13.9	19.2	32.5	42.5	55.5	59.2	58.3	50.0	35.5	26.5	17.0	(22)	
	5%	DB	12.0	11.6	15.4	30.9	44.4	61.8	68.3	63.9	52.3	34.9	22.9	14.3	(23)	
		MCWB	11.1	10.5	14.3	28.5	39.1	52.8	58.0	56.2	47.6	33.2	21.8	13.4	(24)	
	10%	DB	5.7	6.9	9.1	26.0	40.2	56.1	63.9	59.2	48.8	32.6	19.3	9.9	(25)	
		MCWB	4.9	6.0	8.0	24.0	36.3	49.3	56.3	53.6	45.5	31.2	18.2	9.1	(26)	
	0.4%	WB	25.0	19.5	25.5	38.0	47.0	60.4	64.0	62.8	54.4	42.5	31.6	22.2	(27)	
		MCDB	26.4	20.5	27.4	42.8	55.5	74.3	75.2	74.1	62.3	45.9	33.6	23.1	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.0	14.2	19.6	32.3	42.7	56.2	61.0	59.0	50.5	36.0	26.5	17.3	(29)	
		MCDB	18.0	15.3	21.4	34.9	48.7	65.9	70.5	67.4	55.8	37.4	27.6	18.2	(30)	
	5%	WB	11.2	10.6	14.4	28.3	39.3	52.8	58.8	56.5	47.9	33.3	21.7	13.7	(31)	
		MCDB	12.1	11.7	15.7	30.5	43.8	61.2	66.6	63.1	51.4	34.6	22.7	14.4	(32)	
	10%	WB	5.1	6.1	8.4	24.0	36.4	49.3	56.4	54.2	45.9	31.5	18.2	9.3	(33)	
		MCDB	5.7	6.9	9.4	25.8	39.8	55.9	63.5	58.9	48.9	32.7	19.2	10.1	(34)	
	5% DB	MDBR	11.5	12.2	13.6	15.5	12.5	13.6	12.6	10.4	8.2	7.7	10.2	11.4	(35)	
		MCDBR	18.0	15.8	16.9	16.5	16.8	26.0	20.6	18.3	13.9	7.6	10.4	16.1	(36)	
		MCWBR	17.3	14.9	15.6	14.9	11.4	16.3	11.3	9.9	9.0	6.6	10.3	15.7	(37)	
		MCWBR	18.1	15.5	16.8	16.9	16.4	25.0	19.1	16.5	12.9	7.3	10.4	16.3	(38)	
(42) Clear-Sky Solar Irradiance	taub		0.155	0.207	0.249	0.303	0.335	0.361	0.351	0.335	0.300	0.230	0.156	N/A	(40)	
		taud	1.899	2.072	2.211	2.127	2.181	2.313	2.432	2.499	2.515	2.176	1.913	N/A	(41)	
	Ebn at Noon		16	181	249	263	267	263	263	255	234	165	12	0	(42)	
		Edh at Noon	1	13	23	34	37	34	29	24	18	11	1	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		6	135	634	1409	1928	1953	1639	1038	536	157	19	0	(44)	
	RadStd		1	15	25	62	118	143	142	121	57	17	2	0	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	+2.43	+2.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+1.01	+2.12	+2.77	N/A	N/A	N/A	-398	-387	N/A	N/A	

Nomenclature: See separate page